

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Develop a Student Portal Using JavaScript Classes and DOM Manipulation

2. AIM

To design and develop a Student Portal web page that uses an ES6 JavaScript class, form input capture, and DOM manipulation to collect student details and display them dynamically on the page.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and use ES6 class syntax to model a Student object.
 - To capture user input from HTML form fields using `getElementById().value`.
 - To create an object instance and pass it to a function for further processing.
 - To use `innerText` to dynamically update DOM elements with object data.
 - To structure a two-panel layout (input card and display card) using CSS Flexbox.
 - To style form elements, buttons, and cards with CSS3 (border-radius, transitions, hover and focus effects).
 - To build a responsive layout that stacks vertically on smaller screens using a media query.
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4. THEORY

ES6 Classes provide a template for creating objects with shared structure. The `constructor()` method runs when a new instance is created with the `new` keyword, and this refers to the instance being built.

DOM Selection and Manipulation — `document.getElementById()` retrieves a reference to an element by its id; `.value` reads the current text of an input field, and `.innerText` sets the visible text content of an element.

Event Handling — the `onclick` attribute runs a function when a button is clicked; `event.preventDefault()` on the form's `onsubmit` stops the browser's default page-reload behaviour.

CSS Flexbox Layout — `display: flex` on the `.container` arranges the input card and display card side by side, with `gap` controlling the spacing between them.

Responsive Design — a `@media (max-width: 600px)` query changes `flex-direction` to `column` so the two cards stack vertically on narrow screens.

5. ALGORITHM / PROCEDURE

- Create three files: `index.html`, `style.css`, and `script.js`.
 - In `index.html`, link the stylesheet in `<head>` and the script before the closing `</body>` tag.
 - Build a `<main class="container">` holding two `<section class="card">` elements.
 - In the first card, add a form with input fields for Full Name, Roll Number, Department, and CGPA.
 - Add a Submit button that calls `display(createstu())` on click.
 - In the second card, add labelled `<span>` elements with ids to hold the displayed name, roll number, department, and CGPA.
 - In `script.js`, define a `Student` class with a constructor that stores `name`, `rn`, `dept`, and `cgpa`.
 - Write `createstu()` to read all four input values with `getElementById().value` and return a new `Student` object.
 - Write `display(stu)` to set the `innerText` of each output `span` from the given `Student` object's properties.
 - In `style.css`, reset default spacing, set the font stack, and centre the page content.
 - Style `.container` with `display: flex` and `gap`; style `.card` with `padding`, `border-radius`, and a `border`.
 - Style inputs with `padding`, `border-radius`, and a `focus border-colour` transition.
 - Style the button with a dark background, rounded corners, and a `hover background-colour` transition.
 - Add a `@media (max-width: 600px)` rule to stack the cards vertically on narrow screens.
 - Open `index.html` in a browser, fill the form, click Submit, and verify the display card updates.
 - Record expected vs. actual output for each test case.
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6. PROGRAM

`index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>Student Portal</title>
<link rel="stylesheet" href="style.css" />
</head>
<body>

<main class="container">
  <!-- Input Form -->
  <section class="card">
    <h2>Enter Student Details</h2>
    <form id="student-form" onsubmit="event.preventDefault();">
      <div class="field">
        <label for="iname">Full Name</label>
        <input type="text" id="iname" placeholder="e.g. Ramu" />
      </div>

      <div class="field">
        <label for="irn">Roll Number</label>
        <input type="text" id="irn" placeholder="e.g. 101" />
      </div>

      <div class="field">
        <label for="idept">Department</label>
        <input type="text" id="idept" placeholder="e.g. Computer Science" />
      </div>

      <div class="field">
        <label for="icgpa">CGPA</label>
        <input type="number" id="icgpa" step="0.01" min="0" max="10" placeholder="e.g. 8.9"
/>
      </div>

      <button type="button" onclick="display(createstu())">Submit</button>
    </form>
  </section>
```

```
<!-- Output Display -->
<section class="card display-card">
  <h2>Student Details</h2>
  <div class="info-group">
    <span class="label">Name</span>
    <span id="uname" class="value">—</span>
  </div>
  <div class="info-group">
    <span class="label">Roll No</span>
    <span id="urn" class="value">—</span>
  </div>
  <div class="info-group">
    <span class="label">Department</span>
    <span id="udept" class="value">—</span>
  </div>
  <div class="info-group">
    <span class="label">CGPA</span>
    <span id="ucgpa" class="value">—</span>
  </div>
</section>
</main>

<script src="script.js"></script>
</body>
</html>
```

style.css

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
  font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif;
}
```

```
body {
```

```
background-color: #f7f7f8;
color: #1a1a1a;
display: flex;
justify-content: center;
align-items: center;
min-height: 100vh;
padding: 20px;
}
```

```
.container {
  display: flex;
  gap: 24px;
  width: 100%;
  max-width: 720px;
}
```

```
.card {
  flex: 1;
  background: #ffffff;
  padding: 28px;
  border-radius: 12px;
  border: 1px solid #e5e5e7;
}
```

```
h2 {
  font-size: 1.1rem;
  font-weight: 600;
  margin-bottom: 20px;
  color: #111111;
}
```

```
.field {
  margin-bottom: 16px;
}
```

```
label, .label {  
  display: block;  
  font-size: 0.75rem;  
  font-weight: 600;  
  text-transform: uppercase;  
  letter-spacing: 0.05em;  
  color: #6e6e73;  
  margin-bottom: 6px;  
}
```

```
input {  
  width: 100%;  
  padding: 10px 12px;  
  font-size: 0.9rem;  
  border: 1px solid #d2d2d7;  
  border-radius: 6px;  
  outline: none;  
  transition: border-color 0.15s ease;  
}
```

```
input:focus {  
  border-color: #0066cc;  
}
```

```
button {  
  width: 100%;  
  margin-top: 8px;  
  padding: 10px;  
  background-color: #111111;  
  color: #ffffff;  
  font-size: 0.875rem;  
  font-weight: 500;  
  border: none;  
  border-radius: 6px;  
  cursor: pointer;
```

```
    transition: background-color 0.15s ease;
}
```

```
button:hover {
    background-color: #333333;
}
```

```
.display-card {
    background-color: #fafafa;
}
```

```
.info-group {
    margin-bottom: 16px;
    padding-bottom: 10px;
    border-bottom: 1px solid #eeeeef;
}
```

```
.info-group:last-child {
    border-bottom: none;
    margin-bottom: 0;
}
```

```
.value {
    font-size: 0.95rem;
    font-weight: 500;
    color: #1d1d1f;
}
```

```
@media (max-width: 600px) {
    .container {
        flex-direction: column;
    }
}
```

script.js

```
class Student {
```

```

    constructor(name, rn, dept, cgpa) {
        this.name = name;
        this.rn = rn;
        this.dept = dept;
        this.cgpa = cgpa;
    }
}

function createstu() {
    let name = document.getElementById("iname").value;
    let rn = document.getElementById("irn").value;
    let dept = document.getElementById("idept").value;
    let cgpa = document.getElementById("icgpa").value;

    const stu = new Student(name, rn, dept, cgpa);
    return stu;
}

function display(stu) {
    document.getElementById("uname").innerText = stu.name;
    document.getElementById("urn").innerText = stu.rn;
    document.getElementById("udept").innerText = stu.dept;
    document.getElementById("ucgpa").innerText = stu.cgpa;
}

```

7. CODE EXPLANATION

Code	Explanation
class Student	Defines a blueprint object holding name, roll number, department, and CGPA.
constructor()	Initializes each new Student instance with the provided values.
getElementById().value	Reads the current value typed into a form input field.
createstu()	Reads all four form fields and returns a new Student object.
display(stu)	Updates the output spans on the page with fields from the given Student object.

Code	Explanation
onclick="display(createstu())"	Chains input capture and DOM update together in a single click handler.
innerText	Sets the visible text content of a DOM element.
event.preventDefault()	Stops the form's default submission/page-reload behaviour.
display: flex (.container)	Places the input card and display card side by side.
border-radius	Rounds the corners of the cards, inputs, and button.
button: hover	Darkens the button background on mouse-over.
input: focus	Highlights the input border colour when the field is active.
@media (max-width: 600px)	Stacks the two cards vertically on narrow screens.

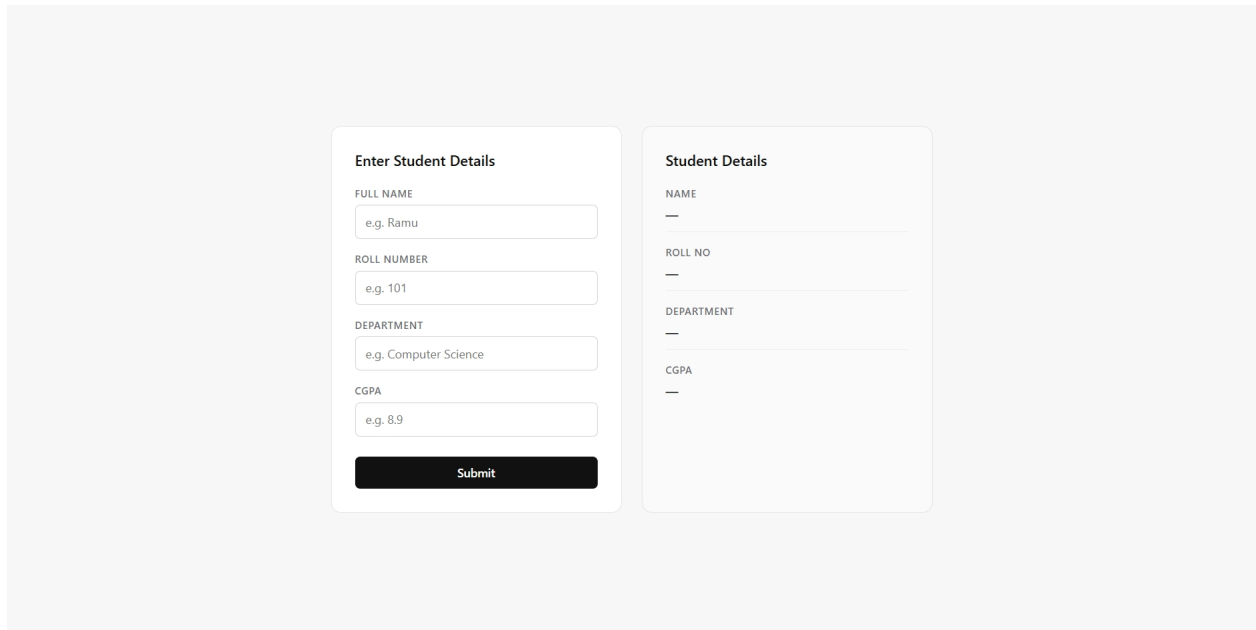
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load the page in a browser	Input card and display card render side by side	Both cards displayed side by side correctly	Pass
TC-02	Leave all fields blank and click Submit	Display card shows blank or default values	Display card fields showed blank text	Pass
TC-03	Enter valid name, roll no, department, and CGPA, then click Submit	Display card updates with the entered values	All four values displayed correctly	Pass
TC-04	Enter a CGPA above 10	Number input restricts or flags the out-of-range value	Browser enforced the max="10" constraint	Pass
TC-05	Click Submit again with different values	Display card updates to show the latest values	Display card refreshed with new values	Pass
TC-06	Focus on an input field	Input border colour changes to blue	Border colour changed on focus	Pass
TC-07	Hover over the Submit button	Button background darkens	Background colour darkened smoothly	Pass
TC-08	Resize the browser to a narrow width	The two cards stack vertically	Cards stacked into a single column	Pass

9. OUTPUT

Output 1: Initial Page Load

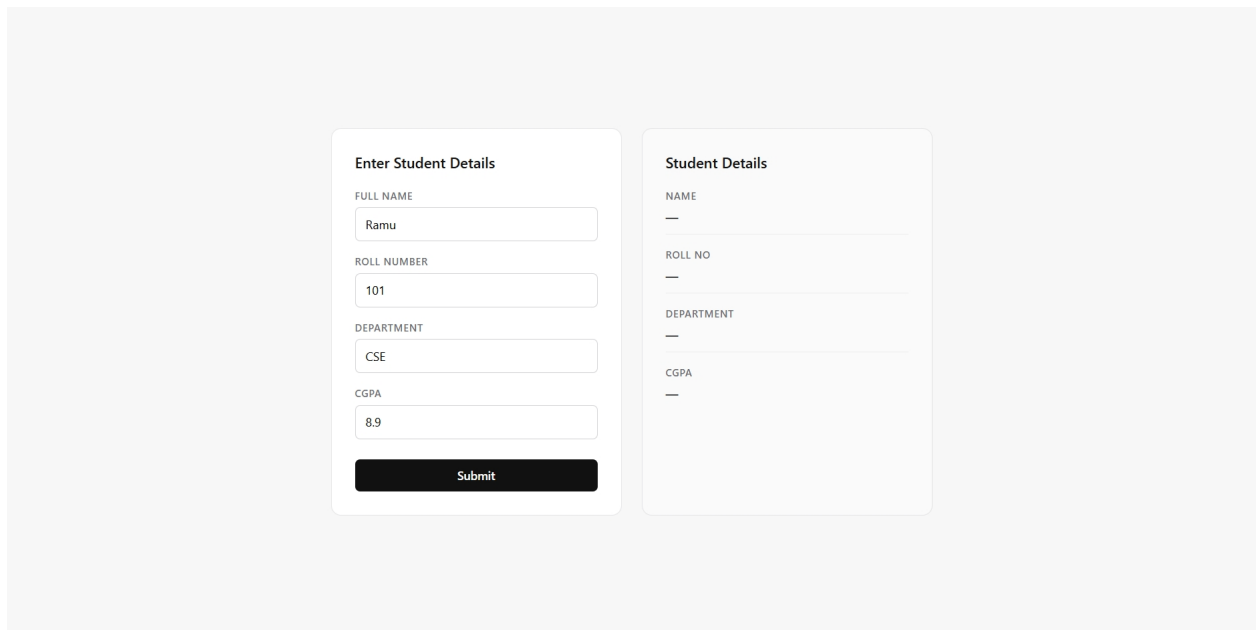
The page renders the input card and the display card side by side. The display card shows a dash (—) placeholder for each field before any data is submitted.



The screenshot shows two side-by-side cards on a light gray background. The left card, titled "Enter Student Details", contains four input fields with placeholder text: "FULL NAME" (e.g. Ramu), "ROLL NUMBER" (e.g. 101), "DEPARTMENT" (e.g. Computer Science), and "CGPA" (e.g. 8.9). A black "Submit" button is at the bottom. The right card, titled "Student Details", shows the same four fields but they are empty, with only horizontal lines indicating where the data would be displayed.

Output 2: Filled Registration Form

The input card shows the Full Name, Roll Number, Department, and CGPA fields filled in with sample student data.



This screenshot is identical to the first one, but the input fields in the "Enter Student Details" card are now filled with sample data: "Ramu" for Full Name, "101" for Roll Number, "CSE" for Department, and "8.9" for CGPA. The "Submit" button remains black.

Output 3: Display Card After Submit

After clicking Submit, the display card updates to show the entered name, roll number, department, and CGPA.

The screenshot displays a web interface for a Student Portal. It features two main components side-by-side on a light gray background. The left component is a form titled "Enter Student Details" with a black "Submit" button at the bottom. The right component is a confirmation card titled "Student Details" showing the entered information.

Form Field	Value
FULL NAME	Ramu
ROLL NUMBER	101
DEPARTMENT	CSE
CGPA	8.9

Student Details

Field	Value
NAME	Ramu
ROLL NO	101
DEPARTMENT	CSE
CGPA	8.9

10. RESULT

Thus, a Student Portal web page was successfully designed and developed using an ES6 JavaScript class, form-based input capture, and DOM manipulation. The page demonstrates object-oriented structuring of data, dynamic display updates, and a responsive two-card Flexbox layout with CSS3 styling. All eight test cases passed successfully, confirming correct data capture, display, and responsive behaviour.